E-commerce Returns & Refunds Optimization

Business Analyst Project

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Domain: Retail / E commerce

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1. Case Study Summary

Domain: Retail / E-commerce

Problem: The company's return rate rose to 22% over two quarters, eroding margins and overwhelming customer service and warehouse teams. Three competing, unconfirmed theories existed for the cause: Marketing suspected aggressive discounting drove impulse-buy returns, Warehouse suspected packaging/damage issues, and there was an unconfirmed suspicion that specific categories and suppliers had disproportionately high return rates.

My Role: *Business Analyst / Data Analyst* : conducted stakeholder analysis across 9 groups, defined data requirements and wrote SQL queries to test all three theories against order and return data, built a cost-benefit case, authored a 5-point gap analysis and full BRD, and defined measurable KPIs directly traceable to the data findings.

Approach: Rather than accepting any stakeholder's theory at face value, this project treated each as a hypothesis to be tested with data using SQL to calculate return rate by discount status, by product category/supplier, and by damage caused. Findings directly shaped both the requirements and a cost-benefit case built for a historically cautious Finance stakeholder.

Key Artifacts: Stakeholder Analysis, Data Requirements & SQL Queries, Cost-Benefit Analysis, As-Is/To-Be Gap Analysis, BRD, KPI Dashboard Visual, User Stories & Acceptance Criteria.

Projected Impact: Return rate reduced from 22% to 10% annually; discount-driven returns from 30% to 16%; damage-driven returns from 15% to 5% worst-category returns from 35% to 20% resellable-condition rate from 60% to 85%. Proposed fix (LKR 3.5M) pays back in under one month against LKR 51M projected annual savings.

2. Stakeholder Analysis

Stakeholder	Interest / Concern	Influence	Support
Customers	Want undamaged products and a smooth, hassle-free return experience if something goes wrong.	Low	Neutral
Customer Service Agents	Want a reduction in return-related complaint volume, which is currently overwhelming their workload.	Medium	Champion
Warehouse Staff	Want fewer returned items overall, and want returns to arrive undamaged/well-packaged so they can be resold.	Medium	Champion
Courier Riders	Want to avoid handling damaged return shipments and want reduced overall return volume.	Low	Cautious
Sellers / Suppliers	Concerned about being implicated if data shows their products/categories drive disproportionate returns, affecting contracts or margins.	Low	Cautious

Marketing Team	Believes discount-driven campaigns may cause impulse-buy returns; wants data to confirm/rule this out, since it could mean rethinking campaign strategy.	High	Champion
Finance Team	Wants precise cost data on returns' impact on margins before approving any proposed fix.	High	Cautious
E-commerce Operations Manager	Accountable for the 22% return rate and under pressure to show measurable improvement.	High	Champion
IT Team	Responsible for building and maintaining dashboards/systems to consolidate and analyze returns data.	Medium	Neutral

3. Data Requirements & SQL Analysis

Three competing theories were tested directly against order and return data, using a simplified 3-table model (Products, Orders, Returns).

3.1 Theory 1: Discount-Driven Returns

```
SELECT
    o.discount_applied,
    COUNT(DISTINCT o.order_id) AS total_orders,
    COUNT(DISTINCT r.return_id) AS total_returns,
    ROUND(COUNT(DISTINCT r.return_id) * 100.0 /
          COUNT(DISTINCT o.order_id), 2) AS return_rate_pct
FROM orders o
LEFT JOIN returns r ON o.order_id = r.order_id
GROUP BY o.discount_applied;
```

3.2 Theory 2: Category/Supplier Concentration

```
SELECT
    p.category,
    COUNT(DISTINCT o.order_id) AS total_orders,
    COUNT(DISTINCT r.return_id) AS total_returns,
    ROUND(COUNT(DISTINCT r.return_id) * 100.0 /
          COUNT(DISTINCT o.order_id), 2) AS return_rate_pct
FROM Products p
JOIN Orders o ON p.product_id = o.product_id
LEFT JOIN Returns r ON o.order_id = r.order_id
GROUP BY p.category
ORDER BY return_rate_pct DESC;
```

3.3 Theory 3: Packaging/Damage

```
SELECT
    p.category,
    COUNT(r.return_id) AS total_returns,
    SUM(CASE WHEN r.damaged = 'Y' THEN 1 ELSE 0 END) AS damaged_returns,
    ROUND(SUM(CASE WHEN r.damaged = 'Y' THEN 1 ELSE 0 END) * 100.0
          / COUNT(r.return_id), 2) AS damaged_pct
FROM Returns r
JOIN Orders o ON r.order_id = o.order_id
JOIN Products p ON o.product_id = p.product_id
GROUP BY p.category
ORDER BY damaged_pct DESC;
```

4. Cost-Benefit Analysis

Item	Estimated Cost / Benefit	Assumption Used
Current cost of returns (annual)	LKR 66,000,000	22% return rate = 26,400 returns/year. Processing: 26,400 x LKR 500 = LKR 13.2M. Lost items (40% non-resellable): 10,560 x LKR 5,000 = LKR 52.8M.
Cost of proposed fix	LKR 3,500,000	Packaging material: LKR 2M. Dashboard build: LKR 1M. Staff training: LKR 0.5M.
Projected annual savings	LKR 51,000,000	Returns reduce to 10% (12,000/yr); processing = LKR 6M. Non-resellable rate falls to 15% (1,800 items); loss = LKR 9M. New total cost = LKR 15M. Savings = LKR 66M - LKR 15M.
Net benefit	LKR 47,500,000	Annual savings (LKR 51M) minus cost of fix (LKR 3.5M).
Payback period	0.8 months	Fix cost (LKR 3.5M) / average monthly savings (LKR 51M / 12).

5. As-Is / To-Be Gap Analysis

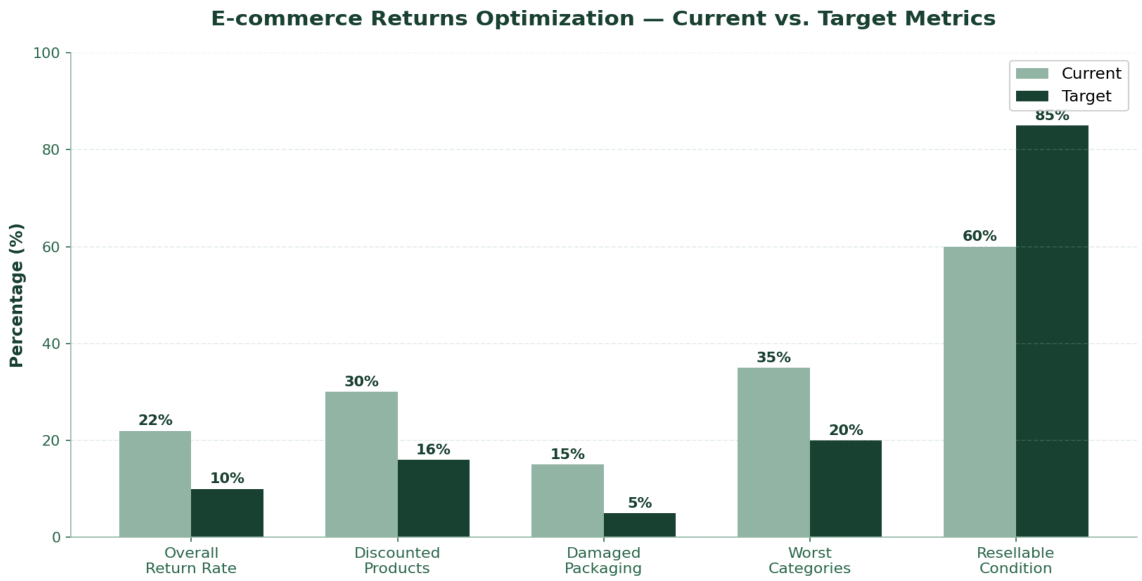
Process Step	As-Is (Current)	To-Be (Future)	Gap
1. Return Reason Capture	No structured field captures why an item was returned.	Every return logged with a standardized return reason category.	Data Model: add return_reason field to Returns table.
2. Discount/Campaign Tracking	No data-backed way to confirm Marketing's discount-return theory.	Return rate by discount status tracked continuously.	New Reporting: recurring dashboard on discount-vs-return-rate query.
3. Packaging & Damage Handling	No standardized packaging quality checkpoint; 15% of returns are damage-related.	Standardized packaging checklist before dispatch.	Process Change: packaging checklist + staff training.
4. Category/Supplier Visibility	Return-rate variance by category/supplier was only a suspicion, never measured.	Return rate by category/supplier reviewed regularly.	New Reporting: recurring category/supplier return-rate report.
5. Returns Cost Visibility (Finance)	Finance had no structured view of the true cost of returns.	Finance has ongoing visibility into returns cost, broken down by cause.	Process Change: institutionalize the cost-benefit calc as a recurring report.

6. Business Requirements Document

Scope: This project covers SQL-based analysis of return drivers, a structured return-reason data model, recurring reporting for Marketing/Operations/Finance, and a cost-benefit case for packaging improvements. Out of scope: campaign strategy redesign, supplier contract renegotiation, and full warehouse system overhaul.

ID	Category	Requirement
BR-01	Data Model	The system shall capture a structured return_reason field for every return, categorized into standard types (damaged, wrong item, changed mind, quality issue).
BR-02	Reporting	The system shall calculate and report return rate segmented by discount status on a recurring basis.
BR-03	Reporting	The system shall calculate and report return rate segmented by product category and supplier on a recurring basis.
BR-04	Reporting	The system shall track the percentage of returned items that are resellable versus non-resellable/damaged.
BR-05	Process	Warehouse packaging shall follow a standardized quality checklist prior to dispatch, tracked against the damage-return rate.
BR-06	Reporting	Finance shall have recurring visibility into the total cost of returns, replacing the previous one-off analysis.

7. KPI Dashboard - Current vs. Target



8. User Stories & Acceptance Criteria

Draft first version, to be reviewed and personalized.

- As a Marketing Manager, I want to see the return rate for discounted vs. non-discounted orders, so that I can determine whether our campaign strategy is contributing to returns.
- As an Operations Manager, I want to see return rates broken down by product category and supplier, so that I can identify which products need quality review.
- As a Warehouse Supervisor, I want to track the percentage of returns caused by damage, so that I can measure whether packaging improvements are working.
- As a Finance Analyst, I want a recurring view of the total cost of returns, so that I don't need to request a manual analysis every time I need this figure.
- As a Customer Service Agent, I want returns logged with a standardized reason code, so that recurring issues can be identified systematically rather than anecdotally.

Acceptance Criteria - Discount Return Rate Story

Given a set of orders within a selected date range,

When the Marketing Manager views the returns dashboard,

Then they see return rate percentages split by discounted and non-discounted orders, updated at least daily.

Acceptance Criteria - Return Reason Capture Story

Given a customer service agent is processing a return, When they complete the return record,

Then they must select a return reason from a standardized list before the return can be finalized.